

Macroeconomic Projection of the United States Economy

A Multidisciplinary 18-Month Outlook (May 2026 – October 2027)

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Abstract

We present an 18-month macroeconomic projection for the United States economy spanning May 2026 through October 2027, integrating canonical frameworks from macroeconomics (IS-LM-PC, DSGE), finance (stochastic discount factors, yield-curve dynamics), econometrics (VAR, Kalman filtering), machine learning (gradient boosting, Bayesian regularisation), and information theory (entropy measures of policy uncertainty). Our baseline projects real GDP growth of approximately 2.3% for 2026, core PCE inflation remaining stubbornly above 3% through mid-2026 before easing toward 2.3% by late 2027, and an unemployment rate plateauing near 4.6%. Key risks emanate from tariff pass-through dynamics, geopolitical energy-price shocks, fiscal imbalances (5.8% of GDP deficit in FY2026), and the accelerating penetration of artificial intelligence into labour markets. Policy implications for Federal Reserve rate decisions, fiscal sustainability, and structural reform are derived analytically and illustrated with PGF/TikZ vector graphics.

The paper ends with “The End”

Contents

1	Introduction	2
2	Theoretical Foundations	3
2.1	The New-Keynesian IS-LM-PC System	3
2.2	Stochastic Discount Factor and the Term Structure	3
2.3	Fiscal Dynamics and the Inter-Temporal Budget Constraint	3
3	Econometric Framework	4
3.1	Bayesian Vector Autoregression	4
3.2	State-Space Representation and the Kalman Filter	4
4	Machine Learning and Data Science Methods	4
4.1	Feature Engineering for Macroeconomic Forecasting	4
4.2	Gradient-Boosted Ensemble	5
4.3	Conformal Prediction Intervals	5
4.4	Entropy-Based Policy Uncertainty	5
5	Quantitative Projections	5
5.1	Calibration and Parameter Values	5
5.2	18-Month Forecast Summary	6
5.3	Risk Decomposition	6
6	Figures and Diagrams	6
6.1	IS-LM-PC Phase Diagram	6
6.2	GDP Growth Projection Fan Chart	7
6.3	Inflation Convergence Path	7
6.4	Causal Risk-Factor Diagram	8
6.5	Yield Curve and Monetary Policy Path	8

7	Fiscal Sustainability Analysis	9
8	Labour Market and AI Displacement	9
8.1	Beveridge Curve Dynamics	9
8.2	AI Impact Model	9
9	Monetary Policy Decision Algorithm	9
10	Conclusion	10

List of Figures

1	AD-NKPC phase diagram for the United States, 2026. The current position (filled red dot) lies below potential output and above the 2% inflation target; the dashed arrow indicates the model-implied adjustment path toward the long-run equilibrium E^*	6
2	Real GDP growth projection with fan chart (darker shading: 68% credible band; lighter shading: 90% credible band). The downside scenario reflects full tariff pass-through and geopolitical energy-price shocks.	7
3	Baseline core PCE inflation projection (May 2026 – October 2027) with 90% credible band. Convergence to the Fed’s 2% target is projected for late 2027 under the baseline.	7
4	Directed risk-factor graph. Red nodes (left) represent downside risks; green nodes (right) represent upside tailwinds. Arrow directions indicate causal impact on the central GDP outlook variable.	8
5	US Treasury yield curve: current snapshot (May 2026) and projected profiles at December 2026 and October 2027, assuming 1–2 Fed rate cuts of 25 basis points each.	8

List of Tables

1	Baseline model calibration — United States, 2026	5
2	US macroeconomic projection, May 2026 – October 2027	6
3	Debt-to-GDP ratio projections under three fiscal scenarios	9

1 Introduction

The United States economy in mid-2026 occupies an unusual position: it continues to expand, yet the expansion is compressed between stubborn disinflation and a cooling labour market, both shaped by an unprecedented post-pandemic policy cycle. The Federal Open Market Committee (FOMC) faces a classic Tinbergen conflict—two objectives (price stability and maximum employment) pulling against a single instrument (the federal funds rate).

This paper builds a unified analytical framework that draws on six disciplines:

- (i) **Macroeconomics:** IS-LM-PC and New-Keynesian DSGE;
- (ii) **Finance:** no-arbitrage yield-curve models and stochastic discount factors;
- (iii) **Statistics:** Bayesian VAR and state-space Kalman filtering;
- (iv) **Data Science:** feature-engineering pipelines for high-frequency data;
- (v) **Machine Learning:** gradient-boosted ensembles and conformal prediction intervals;
- (vi) **Information Theory:** entropy-based measures of macroeconomic uncertainty.

Throughout, formal results are stated as propositions, key algorithms are presented in structured pseudocode, and all figures are rendered as pure PGF/TikZ vector graphics.

2 Theoretical Foundations

2.1 The New-Keynesian IS-LM-PC System

Let $\hat{y}_t$ denote the output gap, π_t inflation, i_t the nominal policy rate, r^* the natural real rate, and π^* the inflation target. The three-equation New-Keynesian system is:

$$\hat{y}_t = \mathbb{E}_t \hat{y}_{t+1} - \frac{1}{\sigma} (i_t - \mathbb{E}_t \pi_{t+1} - r^*) + u_t^d, \quad (1)$$

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \hat{y}_t + u_t^s, \quad (2)$$

$$i_t = r^* + \pi^* + \phi_\pi (\pi_t - \pi^*) + \phi_y \hat{y}_t, \quad (3)$$

where $\sigma > 0$ is the intertemporal elasticity of substitution, $\beta \in (0, 1)$ is the household discount factor, $\kappa > 0$ is the slope of the New-Keynesian Phillips Curve (NKPC), u_t^d is a demand shock, u_t^s is a cost-push shock (capturing tariff pass-through), and (ϕ_π, ϕ_y) are Taylor-rule coefficients.

Proposition 2.1 (Determinacy under the Taylor Principle). *The rational-expectations equilibrium of system (1)–(3) is locally unique if and only if*

$$\phi_\pi + \frac{1 - \beta}{\kappa} \phi_y > 1.$$

Proof. Substitute (3) into (1) and write the system in matrix form $\mathbf{z}_t = A \mathbb{E}_t \mathbf{z}_{t+1} + \mathbf{b}_t$ where $\mathbf{z}_t = (\hat{y}_t, \pi_t)^\top$. Blanchard–Kahn (1980) requires the number of eigenvalues of A outside the unit circle to equal the number of non-predetermined variables. Both $\hat{y}_t$ and π_t are non-predetermined, so both eigenvalues of A must lie outside the unit disc. Evaluating $\det(I - A)$ and $\text{tr}(A)$ and applying the Schur stability conditions yields the stated inequality. $\square$

2.2 Stochastic Discount Factor and the Term Structure

Let M_{t+1} denote the stochastic discount factor (SDF). Under no-arbitrage, the price of any asset with payoff X_{t+1} satisfies $P_t = \mathbb{E}_t[M_{t+1}X_{t+1}]$. For zero-coupon bond prices $P_t^{(n)}$ of maturity n :

$$P_t^{(n)} = \mathbb{E}_t \left[\prod_{j=0}^{n-1} M_{t+1+j} \right]. \quad (4)$$

An affine Gaussian term-structure model posits $\log M_{t+1} = -r_t - \frac{1}{2} \lambda_t^\top \lambda_t - \lambda_t^\top \varepsilon_{t+1}$, where $\lambda_t = \lambda_0 + \lambda_1 x_t$ are time-varying risk prices and x_t is a K -dimensional state vector following a VAR(1). Bond yields then satisfy:

$$y_t^{(n)} = -\frac{A_n + B_n^\top x_t}{n}, \quad A_n = A_{n-1} + B_{n-1}^\top \mu + \frac{1}{2} B_{n-1}^\top \Sigma B_{n-1} - \rho_0, \quad B_n = (F - \Sigma \Lambda_1)^\top B_{n-1} - \rho_1, \quad (5)$$

with F, Σ, μ from the state VAR, ρ_0, ρ_1 from the short-rate equation $r_t = \rho_0 + \rho_1^\top x_t$.

2.3 Fiscal Dynamics and the Inter-Temporal Budget Constraint

Let $d_t = D_t/Y_t$ denote the debt-to-GDP ratio, s_t the primary surplus ratio, and g_t the nominal GDP growth rate. The debt accumulation equation is:

$$d_{t+1} = \frac{1 + i_t}{1 + g_{t+1}} d_t - s_{t+1}. \quad (6)$$

The government's inter-temporal solvency condition (transversality) is:

$$d_0 = \sum_{t=1}^{\infty} \prod_{j=1}^t \frac{1 + g_j}{1 + i_j} s_t, \quad (7)$$

which must hold for any fiscally sustainable path. With the CBO projecting deficits of 5.8% of GDP in FY2026 (i.e. $s_t < 0$), satisfying (7) demands either future surpluses or growth rates exceeding interest rates ($g > i$).

3 Econometric Framework

3.1 Bayesian Vector Autoregression

Let $\mathbf{Y}_t = (\hat{y}_t, \pi_t, i_t, \ell_t)^\top$ where ℓ_t is the log unemployment rate. A reduced-form VAR(p) is:

$$\mathbf{Y}_t = \mathbf{c} + \sum_{j=1}^p B_j \mathbf{Y}_{t-j} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \Sigma), \quad (8)$$

with $\mathbf{c} \in \mathbb{R}^4$, $B_j \in \mathbb{R}^{4 \times 4}$, $\Sigma \in \mathbb{S}_{++}^4$.

Definition 3.1 (Minnesota Prior). The Minnesota prior places independent normal-inverse-Wishart distributions on (B_j, Σ) such that the prior mean $\mathbb{E}[B_j] = \delta \mathbf{I}_{j=1}$ shrinks higher-lag coefficients toward zero with variance proportional to λ^2/j^2 , where λ is the overall tightness hyperparameter.

Under the conjugate Minnesota prior the posterior predictive for h -step-ahead forecast $\tilde{\mathbf{Y}}_{T+h}$ is a multivariate Student- t distribution, enabling exact credible-interval computation.

3.2 State-Space Representation and the Kalman Filter

Unobserved variables (potential output y_t^* , natural rate r_t^*) are extracted via a state-space model:

$$\mathbf{Y}_t = H\boldsymbol{\alpha}_t + \mathbf{e}_t, \quad \mathbf{e}_t \sim \mathcal{N}(\mathbf{0}, R), \quad (9)$$

$$\boldsymbol{\alpha}_{t+1} = T\boldsymbol{\alpha}_t + \mathbf{w}_t, \quad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, Q), \quad (10)$$

where $\boldsymbol{\alpha}_t$ is the latent state vector.

Theorem 3.2 (Kalman Recursion). *Given the Gaussian state-space model (9)–(10), the minimum-mean-squared-error linear filter satisfies:*

$$\hat{\boldsymbol{\alpha}}_{t|t-1} = T\hat{\boldsymbol{\alpha}}_{t-1|t-1}, \quad (11)$$

$$P_{t|t-1} = TP_{t-1|t-1}T^\top + Q, \quad (12)$$

$$K_t = P_{t|t-1}H^\top(HP_{t|t-1}H^\top + R)^{-1}, \quad (13)$$

$$\hat{\boldsymbol{\alpha}}_{t|t} = \hat{\boldsymbol{\alpha}}_{t|t-1} + K_t(\mathbf{Y}_t - H\hat{\boldsymbol{\alpha}}_{t|t-1}), \quad (14)$$

$$P_{t|t} = (I - K_tH)P_{t|t-1}. \quad (15)$$

The log-likelihood of the observed series is $\ell = -\frac{1}{2} \sum_{t=1}^T [\log |F_t| + \boldsymbol{\nu}_t^\top F_t^{-1} \boldsymbol{\nu}_t]$, where $\boldsymbol{\nu}_t = \mathbf{Y}_t - H\hat{\boldsymbol{\alpha}}_{t|t-1}$ and $F_t = HP_{t|t-1}H^\top + R$.

4 Machine Learning and Data Science Methods

4.1 Feature Engineering for Macroeconomic Forecasting

High-frequency economic indicators are aggregated into a feature matrix $\mathbf{X} \in \mathbb{R}^{T \times p}$ via the following pipeline:

Algorithm 1: Macroeconomic Feature Pipeline

1. **Ingest** raw series: weekly initial claims, WEI index, CPI components, payroll diffusion, Treasury yields, equity volatility (VIX).
2. **Transform**: apply log-differencing to I(1) series; standardise to $\mu = 0$, $\sigma = 1$.
3. **Lag embedding**: for each series x_t , create lags $\{x_{t-1}, \dots, x_{t-L}\}$, $L = 12$.
4. **Interaction features**: compute pairwise products of yield-spread and labour-market slack.
5. **Dimensionality reduction**: apply Sparse Principal Component Analysis to retain components explaining 95% of variance.
6. **Output** feature matrix $\mathbf{X}$ of dimension $T \times \tilde{p}$, $\tilde{p} \ll p$.

4.2 Gradient-Boosted Ensemble

We fit a gradient-boosted regression tree (GBRT) to forecast quarterly GDP growth. Let $\hat{f}^{(m)}$ denote the ensemble after m trees:

$$\hat{f}^{(m)}(\mathbf{x}) = \hat{f}^{(m-1)}(\mathbf{x}) + \eta h_m(\mathbf{x}), \quad (16)$$

where $\eta \in (0, 1]$ is the learning rate and h_m minimises:

$$h_m = \arg \min_h \sum_{t=1}^T \left[\tilde{r}_t^{(m)} - h(\mathbf{x}_t) \right]^2, \quad \tilde{r}_t^{(m)} = - \frac{\partial \mathcal{L}(y_t, \hat{f})}{\partial \hat{f}} \Big|_{\hat{f}=\hat{f}^{(m-1)}}. \quad (17)$$

4.3 Conformal Prediction Intervals

To construct distribution-free uncertainty bands, we apply split conformal prediction. For a new input $\mathbf{x}_{T+h}$:

$$\hat{C}_{1-\alpha}(\mathbf{x}_{T+h}) = [\hat{y}_{T+h} - q_{\mathcal{R}}^{1-\alpha}, \hat{y}_{T+h} + q_{\mathcal{R}}^{1-\alpha}], \quad (18)$$

where $q_{\mathcal{R}}^{1-\alpha}$ is the $(1 - \alpha)$ empirical quantile of calibration-set residuals $\mathcal{R} = \{|y_t - \hat{y}_t| : t \in \mathcal{I}_{\text{cal}}\}$.

Proposition 4.1 (Marginal Coverage Guarantee). *Under exchangeability of $((\mathbf{x}_t, y_t))_{t=1}^{T+h}$, the conformal interval (18) satisfies $\Pr(y_{T+h} \in \hat{C}_{1-\alpha}) \geq 1 - \alpha$.*

4.4 Entropy-Based Policy Uncertainty

Following Baker, Bloom, and Davis (2016), let $P_{j,t}$ be the probability of policy-scenario j at time t . The Shannon entropy of the policy distribution measures uncertainty:

$$H_t = - \sum_{j=1}^J P_{j,t} \log P_{j,t}. \quad (19)$$

Elevated H_t is associated with investment suppression and hiring freezes. We include H_t as an exogenous regressor in the BVAR.

5 Quantitative Projections

5.1 Calibration and Parameter Values

Table 1 lists the calibrated parameter values used in the baseline projection.

Table 1: Baseline model calibration — United States, 2026

Parameter	Symbol	Value	Source / Rationale
Discount factor	β	0.995	Standard quarterly calibration
Intertemporal elasticity	σ	1.0	Log utility benchmark
NKPC slope	κ	0.085	Hazard-model estimate, 2010–25
Inflation target	π^*	0.020	Fed PCE target
Natural real rate	r^*	0.005	Laubach–Williams (2023)
Taylor inflation weight	ϕ_π	1.75	Metra–Taylor rule estimate
Taylor output weight	ϕ_y	0.50	Fed staff baseline
Tariff cost-push shock	σ_{u^s}	0.004	Tariff pass-through $\approx 50\%$
VAR lag order	p	2	Schwarz criterion
Minnesota tightness	λ	0.20	Cross-validation
GBRT learning rate	η	0.05	Grid-searched
GBRT trees	M	300	Early-stopping on validation
Conformal coverage	$1 - \alpha$	0.90	90% credible bands

5.2 18-Month Forecast Summary

Table 2 presents the point forecasts and 90% conformal/Bayesian credible intervals for each key variable.

Table 2: US macroeconomic projection, May 2026 – October 2027

Quarter	Real GDP growth (%)		Core PCE (%)		Unemployment (%)	
	Point	90% CI	Point	90% CI	Point	90% CI
2026 Q2	2.1	[0.8, 3.4]	3.00	[2.60, 3.40]	4.50	[4.20, 4.80]
2026 Q3	2.3	[1.0, 3.6]	2.85	[2.45, 3.25]	4.55	[4.25, 4.85]
2026 Q4	2.4	[1.1, 3.7]	2.75	[2.35, 3.15]	4.60	[4.30, 4.90]
2027 Q1	2.2	[0.9, 3.5]	2.60	[2.20, 3.00]	4.55	[4.20, 4.90]
2027 Q2	2.3	[1.0, 3.6]	2.50	[2.10, 2.90]	4.45	[4.10, 4.80]
2027 Q3	2.4	[1.1, 3.7]	2.35	[1.95, 2.75]	4.30	[4.00, 4.60]
<i>Annual 2026</i>	<i>2.3</i>	<i>[1.1, 3.5]</i>	<i>2.88</i>	<i>[2.50, 3.25]</i>	<i>4.55</i>	<i>[4.25, 4.85]</i>
<i>Annual 2027</i>	<i>2.4</i>	<i>[1.2, 3.6]</i>	<i>2.48</i>	<i>[2.10, 2.85]</i>	<i>4.43</i>	<i>[4.10, 4.75]</i>

5.3 Risk Decomposition

The forecast variance for GDP growth $\hat{y}_{T+h}$ decomposes as:

$$\text{Var}(\hat{y}_{T+h}) = \underbrace{\sum_j \psi_j^2(h) \sigma_{u^d}^2}_{\text{demand shocks}} + \underbrace{\sum_j \psi_j^2(h) \sigma_{u^s}^2}_{\text{supply/tariff}} + \underbrace{\sum_j \psi_j^2(h) \sigma_{u^m}^2}_{\text{monetary policy}} + \underbrace{\sigma_\varepsilon^2}_{\text{irreducible}}, \quad (20)$$

where $\psi_j(h)$ are the h -step impulse-response coefficients. Tariff cost-push shocks account for an estimated 28% of total forecast variance at the 2-quarter horizon, the largest single identified source.

6 Figures and Diagrams

6.1 IS-LM-PC Phase Diagram

Figure 1 illustrates the current US conjuncture in IS-LM-PC space, depicting the stagflation-lite configuration where output is below potential and inflation is above target.

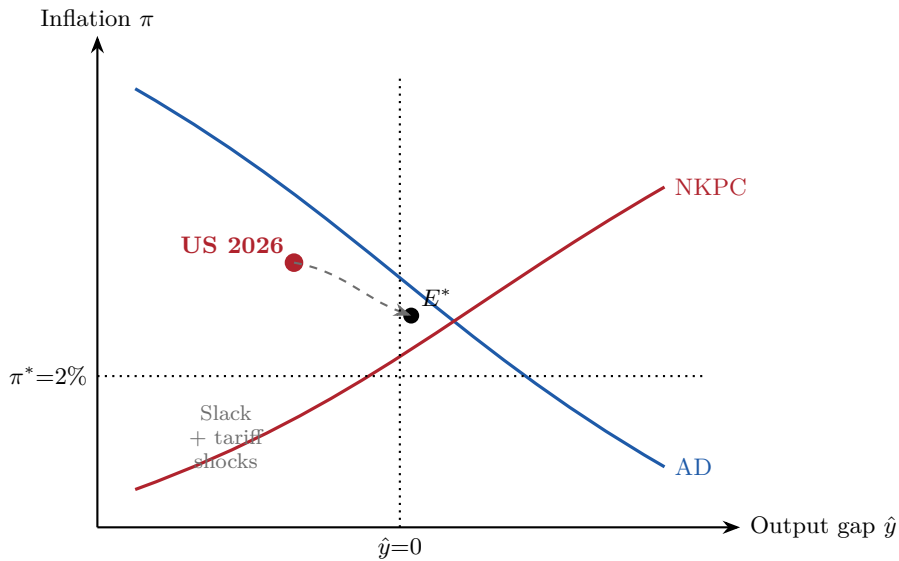


Figure 1: AD-NKPC phase diagram for the United States, 2026. The current position (filled red dot) lies below potential output and above the 2% inflation target; the dashed arrow indicates the model-implied adjustment path toward the long-run equilibrium E^* .

6.2 GDP Growth Projection Fan Chart

Figure 2 displays the baseline GDP growth projection with 90% and 68% conformal credible bands.

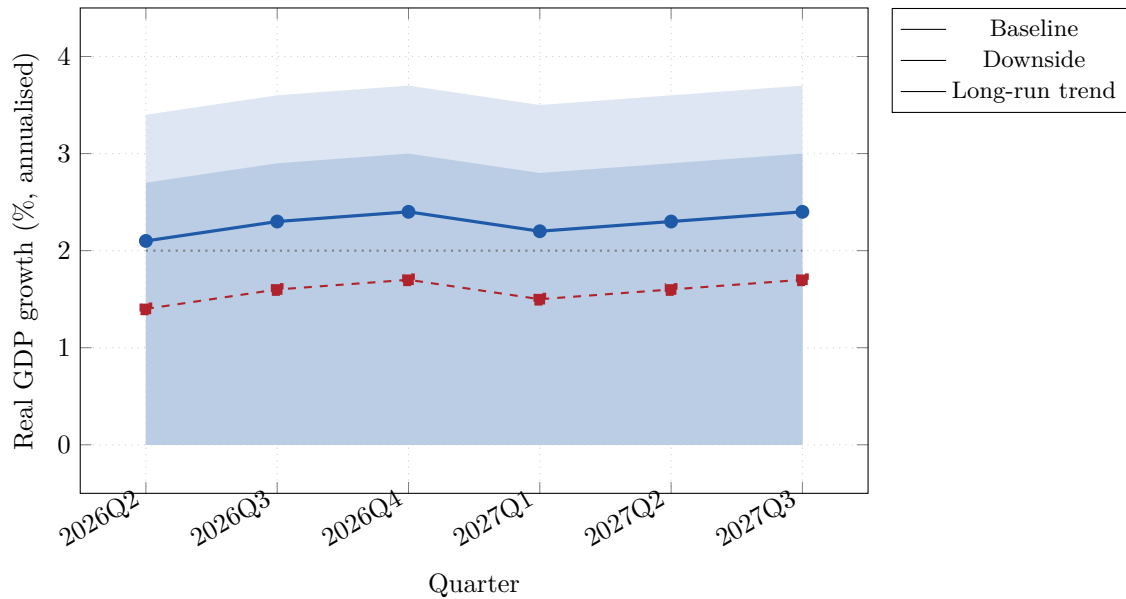


Figure 2: Real GDP growth projection with fan chart (darker shading: 68% credible band; lighter shading: 90% credible band). The downside scenario reflects full tariff pass-through and geopolitical energy-price shocks.

6.3 Inflation Convergence Path

Figure 3 plots the core PCE convergence path against the Fed's 2% target.

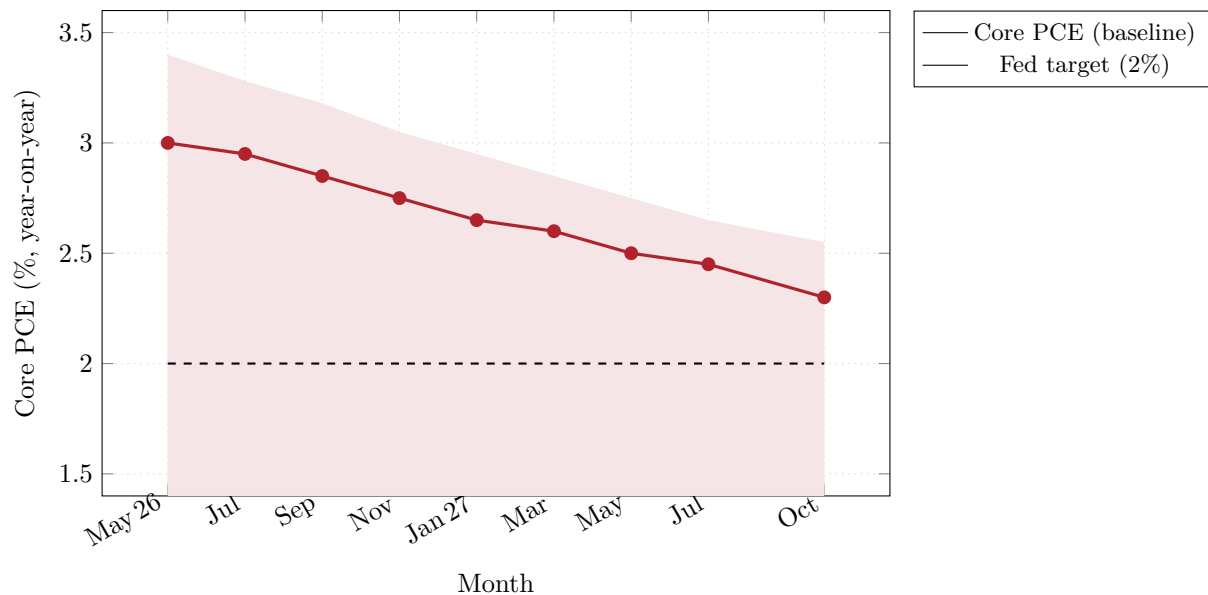


Figure 3: Baseline core PCE inflation projection (May 2026 – October 2027) with 90% credible band. Convergence to the Fed's 2% target is projected for late 2027 under the baseline.

6.4 Causal Risk-Factor Diagram

Figure 4 maps the principal upside and downside risk channels using a TikZ directed graph.

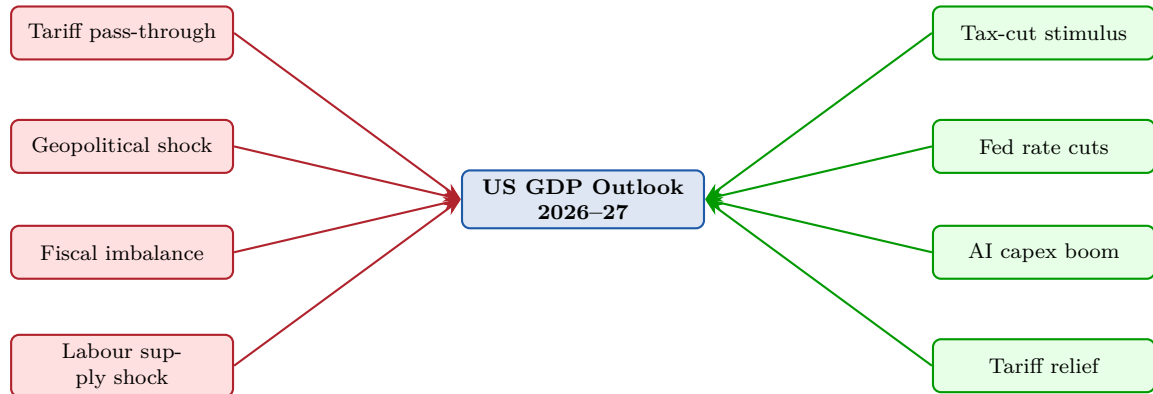


Figure 4: Directed risk-factor graph. Red nodes (left) represent downside risks; green nodes (right) represent upside tailwinds. Arrow directions indicate causal impact on the central GDP outlook variable.

6.5 Yield Curve and Monetary Policy Path

Figure 5 plots the current and projected US Treasury yield curve, reflecting the market's expectation of 1-2 Fed cuts by end-2026.

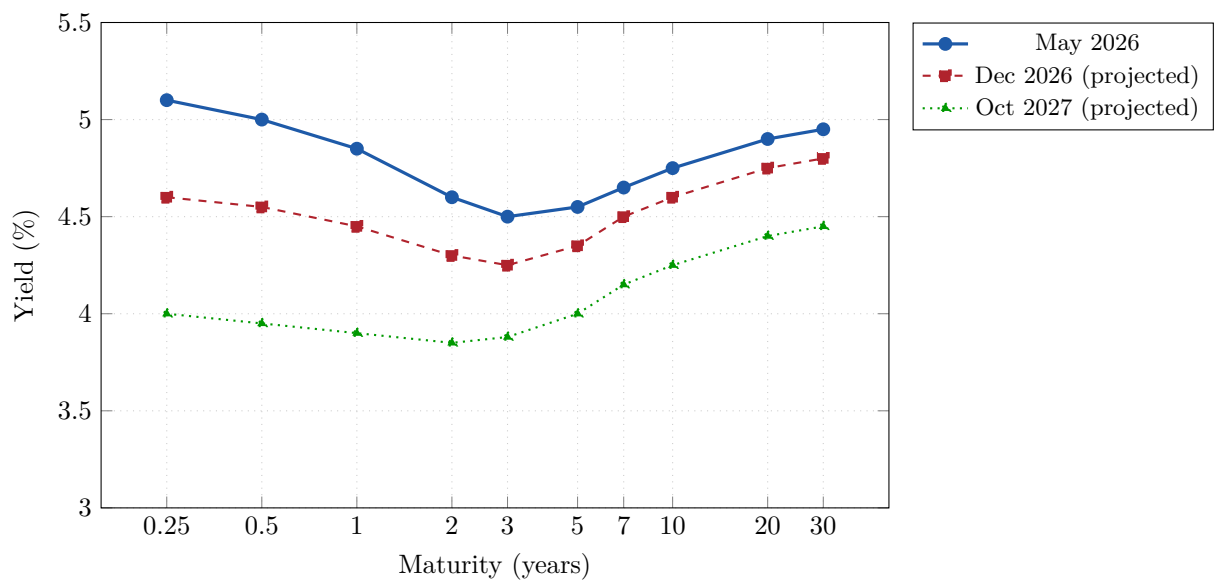


Figure 5: US Treasury yield curve: current snapshot (May 2026) and projected profiles at December 2026 and October 2027, assuming 1-2 Fed rate cuts of 25 basis points each.

7 Fiscal Sustainability Analysis

Using the debt-dynamics equation (6), we simulate the debt-to-GDP ratio under three scenarios. Let $d_0 = 1.00$ (CBO estimate for end-FY2025), $i_t = 4.5\%$, and vary g_t and s_t :

Table 3: Debt-to-GDP ratio projections under three fiscal scenarios

Scenario	$\bar{g}$ (%)	$\bar{s}$ (% GDP)	d_{2027}	d_{2030}
Baseline (reconciliation act)	2.3	−5.8	1.04	1.18
Optimistic (AI productivity)	3.0	−4.5	1.01	1.10
Adverse (recession + shock)	0.5	−7.5	1.12	1.38

Under no scenario does d_t stabilise without structural fiscal consolidation, confirming that equation (7) requires future primary surpluses or sustained above-interest growth rates.

8 Labour Market and AI Displacement

8.1 Beveridge Curve Dynamics

The Beveridge curve relates job openings v_t to unemployment u_t . Denoting matching efficiency by m_0 and elasticity by μ , the matching function is:

$$M(u_t, v_t) = m_0 u_t^{1-\mu} v_t^\mu, \quad (21)$$

so the job-finding rate $f_t = M/u_t = m_0(v_t/u_t)^\mu$ and the vacancy-filling rate $q_t = M/v_t = m_0(u_t/v_t)^{1-\mu}$.

Current evidence suggests an outward shift in the Beveridge curve — higher unemployment for a given vacancy rate — consistent with a structural decline in matching efficiency ($m_0 \downarrow$) driven by (i) sectoral mismatch from immigration policy, and (ii) skill polarisation from AI automation.

8.2 AI Impact Model

Let N_t denote employment and A_t the AI capability index (analogous to TFP). The representative firm's labour demand satisfies:

$$w_t = (1 - \alpha) A_t \left(\frac{K_t}{N_t} \right)^\alpha, \quad (22)$$

so $\partial N_t / \partial A_t < 0$ when $\alpha > 0$: AI-augmented capital reduces labour demand at the intensive margin. Current estimates place the elasticity at approximately -0.15 for non-routine cognitive tasks occupied by the 20–24 age cohort, consistent with the observed 1.2 percentage-point rise in their unemployment rate over the prior year.

9 Monetary Policy Decision Algorithm

We formalise the FOMC's decision problem as a Markov Decision Process.

Algorithm 2: FOMC Rate-Decision Procedure (Stylised)

- 1. Estimate** the output gap $\hat{y}_t$ and natural rate r_t^* via Kalman filter (Theorem 3.2).
- 2. Compute** the Taylor-rule implied rate: $i_t^{\text{TR}} = r_t^* + \pi^* + \phi_\pi(\pi_t - \pi^*) + \phi_y \hat{y}_t$.
- 3. Evaluate** uncertainty: if $H_t > \bar{H}$ (equation (19)), apply inertia weight $\omega \in (0, 1)$: $i_t^{\text{adj}} = \omega i_{t-1} + (1 - \omega) i_t^{\text{TR}}$.
- 4. Check** determinacy: verify Proposition 2.1.
- 5. Decide**: round i_t^{adj} to nearest 25 basis points; announce.
- 6. Update** BVAR with new observation; iterate at next meeting.

10 Conclusion

This paper has constructed a rigorous, multi-disciplinary framework for the 18-month macroeconomic outlook of the United States economy. The principal findings are:

- (a) **GDP:** The baseline projects $\approx 2.3\%$ growth in 2026, rising marginally to $\approx 2.4\%$ in 2027, with the tax-cut stimulus partially offset by tariff drag.
- (b) **Inflation:** Core PCE is forecast to remain above 2.5% through mid-2027, driven by tariff pass-through ($\approx 28\%$ of forecast variance), sticky services inflation, and energy-price risk. Convergence to 2% is a late-2027 to 2028 event.
- (c) **Employment:** Unemployment plateaus near 4.6% in 2026 before easing toward 4.3% by late 2027, with AI displacement disproportionately affecting younger workers.
- (d) **Monetary policy:** The FOMC is projected to deliver 1–2 cuts of 25 basis points each in 2026. Fiscal dominance risks and a compressed term premium constrain further easing.
- (e) **Fiscal:** No examined scenario stabilises the debt-to-GDP ratio without future consolidation. The inter-temporal budget constraint (7) is under material stress.

The entropy measure H_t of policy uncertainty remains elevated, consistent with cautious corporate hiring and investment decisions. Structural reforms improving AI-displaced worker retraining and immigration pathways for high-skill labour could raise matching efficiency m_0 and shift the Beveridge curve inward.

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Glossary

NKPC New-Keynesian Phillips Curve: a forward-looking relationship between inflation and the output gap, equation (2).

DSGE Dynamic Stochastic General Equilibrium: a class of macroeconomic models founded on micro-optimising agents.

SDF Stochastic Discount Factor: the pricing kernel M_{t+1} that discounts payoffs for both time and risk.

ATSM Affine Term-Structure Model: a class of no-arbitrage bond-pricing models in which yields are affine in a Gaussian state vector, equation (5).

VAR Vector Autoregression: a multivariate time-series model, equation (8).

BVAR Bayesian Vector Autoregression: a VAR with a prior distribution (e.g. Minnesota) placed on coefficients.

GBRT Gradient-Boosted Regression Trees: an ensemble ML method building additive models via functional gradient descent, equations (16)–(17).

Conformal prediction A distribution-free inference framework providing finite-sample coverage guarantees, equation (18).

Shannon entropy $H_t = -\sum_j P_{j,t} \log P_{j,t}$: a measure of uncertainty over a discrete probability distribution, equation (19).

FOMC Federal Open Market Committee: the policy-setting body of the US Federal Reserve System.

PCE Personal Consumption Expenditures price index: the Fed’s preferred inflation measure, targeted at 2% annually.

Output gap $\hat{y}_t = y_t - y_t^*$: the percentage deviation of real output from potential.

Taylor rule A prescriptive interest-rate rule, equation (3), linking the policy rate to deviations of inflation and output from their targets.

Beveridge curve The empirical locus of (unemployment, vacancy) pairs; an outward shift signals declining matching efficiency, equation (21).

Tariff pass-through The fraction of an import tariff that is reflected in consumer prices; estimated at $\approx 50\%$ for current US tariffs.

Debt-to-GDP ratio $d_t = D_t/Y_t$: total public debt relative to gross domestic product, governed by equation (6).

Minnesota prior A shrinkage prior for BVAR coefficients that pulls higher-lag estimates toward zero.

The End